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Safety-Critical Passive Control for Robotic Soft Tissue Interaction via Control Barrier Functions

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***Abstract*—In robotic physical interaction tasks, passive control is significant to guarantee robust stability of the systems with unknown environments. In this article, a safety-critical passive control approach via control barrier functions (CBFs) is proposed for robotic soft tissue cutting tasks. To achieve stable cutting motion, this approach builds upon unified force-impedance control enhanced with the energy tanks that constrain the system energy. The energy tanks are formulated with optimization theory to constrain both time-varying energy and power flow, incorporated with high-order time-varying CBFs (HoTvCBFs). To address unknown higher order derivatives in HoTvCBFs under external disturbances, a time-varying projection-to-state safe differentiator-adaptive HoTvCBF (tPSSf-DA–HoTvCBF) approach is developed. The proposed approach employs an input-to-state stable (ISS) differentiator to estimate unknown parameters with bounded errors, and adaptively adjusts safety margins to reduce conservatism compared to fixed-bound approaches. Furthermore, to address potential conflicts between the time-varying CBF constraints and the input constraints, a dynamic-convergence approach, which transforms decay rates into optimization variables to ensure solution feasibility, is introduced. A series of robotic soft tissue cutting experiments is carried out to validate the effectiveness of the proposed approach, demonstrating the capability to handle time-varying constraints and maintain robustness.**

***Index Terms*—Energy tanks, passivity, safe manipulation.**

I. Introduction

ROBOTIC soft tissue interaction involves complex motion transitions. Fig. 1 illustrates a typical example of robotic interaction: a robotic manipulator performing soft tissue cutting operations that involve transitioning between free-space operation and soft tissue contact, namely, noncontact/contact transitions [1]. During these transitions, the manipulator experiences discontinuous dynamics, leading to large impact forces, contact instability, or even system oscillations [2]. To guarantee system stability, passive control is introduced [3] to manage energy exchange, particularly effective when interacting with unknown environments [4], [5]. Therefore, a safety-critical passive controller is essential for stable and safe cutting task execution.

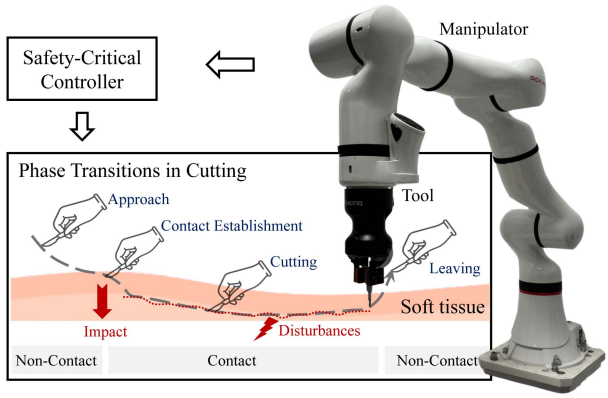


Fig. 1. Typical tool-environment interaction application, i.e., a soft tissue cutting task. The system consists of a robotic manipulator with tactile sensors mounted on the gripper to measure interaction forces at the cutting interface. A safety-critical controller utilizing energy tank management ensures safe and stable operation under contact/noncontact transitions and unpredictable disturbance.

The safe physical interaction depends on robot compliance to dynamically adapt to environmental changes [6], which has been widely adopted in cutting tasks [7], [8], [9]. To manage both safe motion and precise force tracking, unified force-impedance control (UFIC) has been presented in [10], which integrates impedance regulation with direct force control. Considering the system passivity requirement, the stability of UFIC can be analyzed through energy tanks [11]. In this study, the energy tanks implement a virtual energy storage mechanism to regulate the energy budget for control actions [12], [13]. Typically, UFIC employs a valve-based design where the energy tanks are activated or deactivated via Boolean logic [10], [14]. Nevertheless, the discrete "if–else" switching strategy produces nonsmooth control commands, causing the chattering that increases mechanical wear on robotic actuators [15]. In addition, since a

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large amount of energy might be drawn rapidly with abrupt robotic action, ensuring passivity through energy tanks does not inherently guarantee safety [16], which indicates that only limiting energy is insufficient, and the constraints on the rate of energy transfer, i.e., power flow, are also required.

Control barrier functions (CBFs) offer an alternative methodology to enforce the safety constraints continuously, ensuring that the system operates within defined boundaries and generates smooth control commands [17], [18]. As pointed from [19], when the constraints are applied to preserve passivity by altering desired control actions, the operation performance often degrades. In order to minimize this performance reduction, CBFs limit modifications to the control actions as much as possible to prevent unstable system behavior [20]. When control inputs indirectly influence safety constraints, High-order CBFs (HoCBFs) are proposed in [21] for systems with arbitrary relative degree to guarantee safety. The work presented in [22] implements HoCBFs in the energy management systems, simultaneously considering energy constraints and power flow constraints into a unified HoCBF quadratic program (QP) optimization problem, preventing system damage from rapid energy extraction or injection. When power limits remain fixed during task execution, robots may fail to achieve control objectives if the limit is set too conservatively, or experience instability if set too aggressively. Thus, multiphase cutting tasks exhibit varying power and energy demands across different phases. To accommodate these varying demands, phase-specific power constraints that adjust dynamically according to task requirements have been proposed [23]. Under such time-varying constraints, standard time-invariant CBF approach cannot maintain constraint satisfaction, because this approach do not account for the time-derivative of the changing constraint boundary.

In this article, HoTvCBFs are integrated within the energy tank optimization. In practice, the safety guarantees provided by CBFs rely on precise system models and are compromised by model uncertainties and disturbances [24]. To quantify the effect of disturbances on safety guarantees, input-to-state safety (ISSf) has been proposed in [25]. Nevertheless, ISSf is commonly used for compensating the matched disturbances entering through the control input channel, which is not suitable for unmatched disturbances acting directly on the system states. To address this limitation, projection-to-state safety (PSSf) quantifies safety through the projected disturbance acting on the barrier function derivative [26]. However, the existing PSSf formulations, such as the study presented in [27], apply only to relative-degree-one CBFs. For HoCBFs, the safety constraint involves higher order derivatives of the barrier function, which depend on unknown time-varying disturbances and cannot be directly bounded.

This article proposes a time-varying PSSf differentiator-adaptive HoTvCBF (tPSSf-DA–HoTvCBF) approach. To estimate unknown higher order derivatives, the differentiator-adaptive (DA) component employs an input-to-state stable (ISS) differentiator. Through bounding estimation errors, the ISS differentiator avoids noise amplification inherent in numerical differentiation. To adaptively bound the estimation error, time-varying PSSf (tPSSf) is utilized, reducing conservatism compared to standard robust CBF methods, such as the study presented in [26], that rely on static worst-case bounds. In fact, when energy tank constraints conflict with input constraints, the QP optimization problem may become infeasible [28]. To guarantee feasibility under such conflicts, a dynamic-convergence form of HoTvCBF-QP is proposed. This formulation transforms the fixed decay rate parameters into optimization variables, enabling adaptive constraint relaxation. Finally, the validation of the proposed approach is proved through a robotic soft tissue cutting task and the experimental platform is illustrated in Fig. 1. It should be mentioned that several tactile sensors are utilized in the robotic prototype as sensing feedback. The key advantage of using tactile sensing is that it enables the system to estimate tool localization and tool-environment forces through gripper-handle contact pattern analysis, thereby allowing adaptation to various tools without requiring modifications to the setup. More details can be found in our previous work [29]. The main contributions of this work are summarized as follows.

1) Different with time-invariant CBF hard to maintain time-varying constraint satisfaction, this article proposes an energy tank optimization formulation incorporated with time-varying CBFs and HoTvCBFs to ensure smooth transitions in multiphase cutting tasks with time-varying power requirements.
2) To handle unmatched disturbances with unknown derivatives in practical cutting tasks, the tPSSf approach for HoTvCBFs is integrated with an ISS differentiator that estimates the unknown propagation dynamics and bounds estimation errors, thereby ensuring robust safety.
3) Considering the possible conflicts between time-varying energy tank constraints and input constraints, a dynamic-convergence form of HoTvCBF-QP is utilized to guarantee feasible optimal solutions.

The rest of this article is organized as follows. Section II reviews the preliminaries. Section III presents the stability analysis of the cutting process and formulates the problem. Section IV introduces the proposed methodology. Section V presents the experimental results. Finally, Section VI concludes this article.

II. Preliminaries

A. Passivity and Energy Tanks

The passivity theory can guarantee stability in unknown environments [30]. Consider a dynamical system with state ξ, input u, and output y. The system is passive if there exists a continuously differentiable positive semidefinite function $S(\xi)$, namely, the storage function, satisfying

$$S(\xi(t_1)) - S(\xi(t_0)) \leq \int_{t_0}^{t_1} y^T(t)u(t)dt \tag{1}$$

for all $t_1 \geq t_0$ and admissible inputs $u(t)$.

The system stability is guaranteed when input u and output y satisfy

$$\dot{S}(\xi(t_1)) - \dot{S}(\xi(t_0)) \leq u^T y. \tag{2}$$

To maintain system passivity, an energy tank functions as a virtual reservoir regulating both energy and power flow between controller and system [23]. Here, T_t represents stored energy,

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and $P_t \leq 0$ denotes power flow rate, which evolves as

$$\dot{T}_t = \alpha P_t, \quad \alpha = \begin{cases} 0 & \text{if } P_t \leq 0 \text{ and } T_t \leq E_{\min} \\ 1 & \text{otherwise} \end{cases} \tag{3}$$

with power regulation

$$\alpha^\dagger = \begin{cases} \frac{P_{\min}}{P_t} & \text{if } P_t < P_{\min} \leq 0 \\ 1 & \text{otherwise.} \end{cases} \tag{4}$$

Here, $P_t \leq 0$ indicates energy dissipation, $E_{\min}$ and $P_{\min}$ are the minimum allowable energy and power, respectively. The control law, considering both energy and power constraints, is given by

$$P_t^+ = \alpha^\dagger \alpha P_t. \tag{5}$$

B. Control Barrier Functions

For controller design, the second-order system can be reformulated into control affine form

$$\dot{\xi} = f(\xi) + g(\xi)u \tag{6}$$

where $\xi \in \mathbb{R}^n$ denotes the system state, $u \in \mathbb{R}^m$ represents the control input, $f(\xi)$ and $g(\xi)$ are the drift and input dynamics.

Consider a continuously differentiable function $h : \mathbb{R}^n \times [t_0, \infty) \to \mathbb{R}$ that defines a safe set

$$\mathcal{C} = \{\xi \in \mathbb{R}^n | h(\xi) \geq 0\} \tag{7}$$

with boundary $\partial\mathcal{C} = \{\xi \in \mathbb{R}^n | h(\xi) = 0\}$ and interior $Int(\mathcal{C}) = \{\xi \in \mathbb{R}^n | h(\xi) >\}$. To maintain $\xi(t) \in \mathcal{C}$, CBF is introduced as follows.

Definition 1 (CBFs [31]): For a dynamical system in (6) with f and g being locally Lipschitz continuous, a continuously differentiable function $h : \mathbb{R}^n \times [t_0, \infty) \to \mathbb{R}$ is a time-varying CBF if there exists a class $\mathcal{K}$ function α such that

$$\sup_{u \in \mathbb{R}^m} [L_f h(\xi) + L_g h(\xi) u] > -\alpha(h(\xi)). \tag{8}$$

However, when $L_g h(\xi)$ vanishes, higher order derivatives of h must be considered, leading to the concepts of relative degree and HoCBFs.

Definition 2 (Relative degree [32]): For systems with high-relative degree constraints, an mth-order differentiable function $h(\xi)$ is characterized by

$$\psi_0 = h(\xi) \tag{9}$$

$$\psi_i = \dot{\psi}_{i-1} + \alpha_i(\psi_{i-1}), i \in \{1, \ldots, m\} \tag{10}$$

where $\alpha_i(\cdot)$ are differentiable class $\mathcal{K}$ functions. The relative degree m indicates that the control input u first appears explicitly in the mth time derivative of h. When $m = 1$, the HoCBF reduces to the first-order case in (8).

Definition 3 (Time-invariant HoCBFs [21]): A function $h : \mathbb{R}^n \to \mathbb{R}$ represents a HoCBF with relative degree m for the system (6) when differentiable class $\mathcal{K}$ functions $\alpha_i, i \in \{1, \ldots, m\}$ exist such that

$$\sup_{u \in \mathcal{U}} \left[L_f^m h(\xi) + L_g L_f^{m-1} h(\xi) u + O(h(\xi)) \right] \geq -\alpha_m(\psi_{m-1}(\xi)) \tag{11}$$

where the term $O(h(\xi))$ captures the remaining Lie derivatives along f

$$O(h(\xi)) = \sum_{j=1}^{m-1} L_f^j(\alpha_{m-j}(\psi_{m-j-1}))(\xi). \tag{12}$$

Remark 1: In existing literature, the HoCBFs are commonly utilized to enforce safety constraints that are assumed to be time-invariant [21]. Consequently, the explicit time-dependent terms are omitted in Definition 3. Although effective for static environments, this simplification can not handle time-varying constraints.

III. Stability Analysis for Cutting

A. System Dynamics

As shown in Fig. 1, the tool-tissue contact interaction is modeled as a nonlinear viscoelastic model

$$F_e = K_e(-\delta)^a - \zeta(-\delta)^b \dot{\delta} \tag{13}$$

where δ represents the penetration depth, K_e and ζ are the stiffness and damping coefficients, a and b are nonlinear parameters.

The dynamics of an n-joint manipulator are formulated in joint space as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} = \tau + J^\top(q) F_e \tag{14}$$

and in Cartesian space as

$$M_c(q)\ddot{x} + C_c(q, \dot{q})\dot{x} = F_{ctl} + F_e \tag{15}$$

where $q \in \mathbb{R}^n$ represents the joint position, $x \in \mathbb{R}^{n_f}$ denotes the end-effector position, τ is the joint torque, F_{ctl} is the control input, F_e is the external force, and M, C, and J denote the inertia, Coriolis, and Jacobian matrices, respectively.

B. Unified Force-Impedance Control

To achieve simultaneous motion and force regulation, the UFIC strategy structures the control input F_{ctl} as a combination of impedance F_{imp} and force F_F control components

$$F_{\text{ctl}} = F_{\text{imp}} + F_F. \tag{16}$$

Defining the position error $\tilde{x} = x_d - x$ and the force tracking error $e_f = F_e - F_d$, the control terms are designed as

$$F_{\text{imp}} = M_c \ddot{x}_d + C_c \dot{x}_d + K_c \tilde{x} + D_c \dot{\tilde{x}}$$

$$F_F = K_p e_f + K_d \dot{e}_f + K_i \int_0^t e_f(t) dt - F_d \tag{17}$$

where K_c and D_c are the impedance gains, and K_p, K_d, and K_i are the PID gains for force control. Substituting (16) and (17) into (15) yields the closed-loop error dynamics

$$M_c \ddot{\tilde{x}} + (C_c + D_c)\dot{\tilde{x}} + K_c \tilde{x} + F_F + F_e = 0. \tag{18}$$

C. Nominal Dynamics of Energy Tank

For system stability analysis, a storage function is defined as

$$S(\tilde{x}, \dot{\tilde{x}}) = \frac{1}{2}\dot{\tilde{x}}^T M_c \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^T K_c \tilde{x} \tag{19}$$

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and its time derivative is

$$\begin{aligned}\dot{S}(\tilde{x},\dot{\tilde{x}}) &= -\dot{\tilde{x}}^T D_c \dot{\tilde{x}} + \dot{x}^T F_F + \dot{x}^T F_e - \dot{x_d}^T (F_F + F_e) \\ &\leq \dot{x}^T F_F + \dot{x}^T F_e - \dot{x_d}^T (F_F + F_e) \end{aligned} \tag{20}$$

where $\dot{M}_c - 2C_c$ is skew-symmetric, and D_c is the positive-definiteness of the damping matrix. Energy exchange between the system and the environment relies on the passivity with respect to the port $(\dot{x}, F_F)$. The instantaneous power associated with the nonpassive component is defined as

$$P_t = \dot{x}^T F_F \tag{21}$$

where P_t represents the energy flow introduced by F_F. By ensuring F_F satisfies this passivity condition, the term $\dot{x}^T F_F$ dissipates energy, preventing unbounded growth of the system's total energy, represented by the storage function $S(\tilde{x}, \dot{\tilde{x}})$.

To manage the power flow during interaction, the energy tank is modeled as

$$\ddot{e}_f = k_f (P_t - \dot{e}_f) \tag{22}$$

where e_f represents the energy level in the tank, $\dot{e}_f$ tracks the actual power flow, and $k_f > 0$ is the control gain.

For controller design, this second-order system can be reformulated into control affine form (6) with

$$\xi = \begin{bmatrix} e \\ \dot{e} \end{bmatrix}, \quad f(\xi) = \begin{bmatrix} \dot{e} \\ -k_f \dot{e} \end{bmatrix}, \quad g(\xi) = \begin{bmatrix} 0 \\ k_f \end{bmatrix}, \quad u = P_t. \tag{23}$$

D. Problem Formulation

However, real robotic systems inevitably experience disturbances. The true system dynamics are described as

$$\dot{\xi} = f(\xi) + g(\xi)u + \tilde{d}(t) \tag{24}$$

where $\tilde{d}(t) : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ represents the unmatched disturbance acting directly on system states. In robotic cutting tasks, such disturbances arise from unmodeled contact dynamics, sensor noise, and model uncertainty.

To ensure passivity and prevent power/energy violations, the energy tank need to satisfy time-varying safety constraints

$$\begin{aligned} \text{(Power Limits)} \quad & h_1(\xi,t) = \dot{e} - P_{\min}(t) \geq 0 \\ & h_2(\xi,t) = P_{\max}(t) - \dot{e} \geq 0 \\ \text{(Energy Limits)} \quad & h_3(\xi,t) = e - E_{\min}(t) \geq 0 \\ & h_4(\xi,t) = E_{\max}(t) - e \geq 0 \end{aligned} \tag{25}$$

where $P_{\min}(t)$ and $P_{\max}(t)$ denote time-varying power bounds, $E_{\min}(t)$ and $E_{\max}(t)$ are time-varying energy bounds. The safe set is defined as

$$\mathcal{C}(t) = \bigcap_{i=1}^{4} \xi : h_i(t,\xi) \geq 0. \tag{26}$$

Classical energy tanks in [33] enforce these constraints via discrete switching

$$\beta = \begin{cases} 0 & \text{if } h_i < 0 \text{ or energy depleted} \\ 1 & \text{otherwise} \end{cases}$$
$$u = \beta \cdot u_{\text{nom}} \tag{27}$$

where $\beta \in \{0, 1\}$ is the binary scaling factor and $u_{\text{nom}} = P_t$ is the nominal control input. This leads to discontinuous control signals and chattering [34]. CBFs provide an alternative by solving

$$\begin{aligned} u^* = \arg\min_{u \in \mathbb{R}} \quad & \frac{1}{2}\|u - u_{\text{nom}}\|^2 \\ \text{s.t.} \quad & \dot{h}_i(\xi,t,u) + \alpha_i h_i(\xi,t) \geq 0, \quad i = 1,\ldots,4 \end{aligned} \tag{28}$$

which guarantees $h_i(\xi,t) \geq 0$ for all $t \geq 0$ continuously.

However, applying standard HoCBF theory in (11) to time-varying constraints (25) under disturbed dynamics (24) presents three challenges. First, computing time derivatives $\dot{P}_{\min}(t)$ and $\ddot{E}_{\min}(t)$ amplifies measurement noise and may cause safety violations. Second, unmatched disturbances $\tilde{d}(t)$ directly corrupt CBF derivatives, invalidating nominal safety guarantees. Finally, when energy tank constraints conflict with input constraints, the QP optimization problem may become infeasible.

As illustrated in Fig. 2, this article develops a robust safety approach that extends HoCBFs theory to time-varying constraints, incorporates tPSSf to bound the effect of unmatched disturbances via time-varying robust margins, and employs ISS high-gain observers to robustly estimate noisy derivatives.

IV. Methodology

A. Energy Tank Formulation With HoTvCBF Constraints

In (25), the HoCBF theory assumes time-invariant constraints and thus cannot directly handle the time-varying safety requirements. This section extends the HoCBF theory to accommodate explicit time dependence in barrier functions.

Definition 4 (HoTvCBFs): Consider a continuously differentiable function $h : \mathbb{R}^n \times [t_0, \infty) \to \mathbb{R}$ with relative degree m with respect to the system. Define a sequence of auxiliary functions $\psi_i : \mathbb{R}^n \times [t_0, \infty) \to \mathbb{R}$ as

$$\begin{aligned} &\psi_0(\xi,t) \triangleq h(\xi,t) \\ &\psi_i(\xi,t) \triangleq \dot{\psi}_{i-1}(\xi,t) + \alpha_i(\psi_{i-1}(\xi,t)), \quad i = 1,\ldots,m-1 \end{aligned} \tag{29}$$

where α_i are differentiable class-$\mathcal{K}$ functions. The function h is a HoTvCBF if there exists an $\alpha_m \in \mathcal{K}$ such that for all $(\xi,t) \in \bigcap_{i=1}^{m}\{\psi_{i-1}(\xi,t) \geq 0\} \times [t_0, \infty)$

$$\begin{aligned} &\sup_{u \in \mathcal{U}} \left[L_f^m h(\xi,t) + L_g L_f^{m-1} h(\xi,t) u + \frac{\partial^m h(\xi,t)}{\partial t^m} + O(h(\xi,t)) \right] \\ &\geq -\alpha_m(\psi_{m-1}(\xi,t)) \end{aligned} \tag{30}$$

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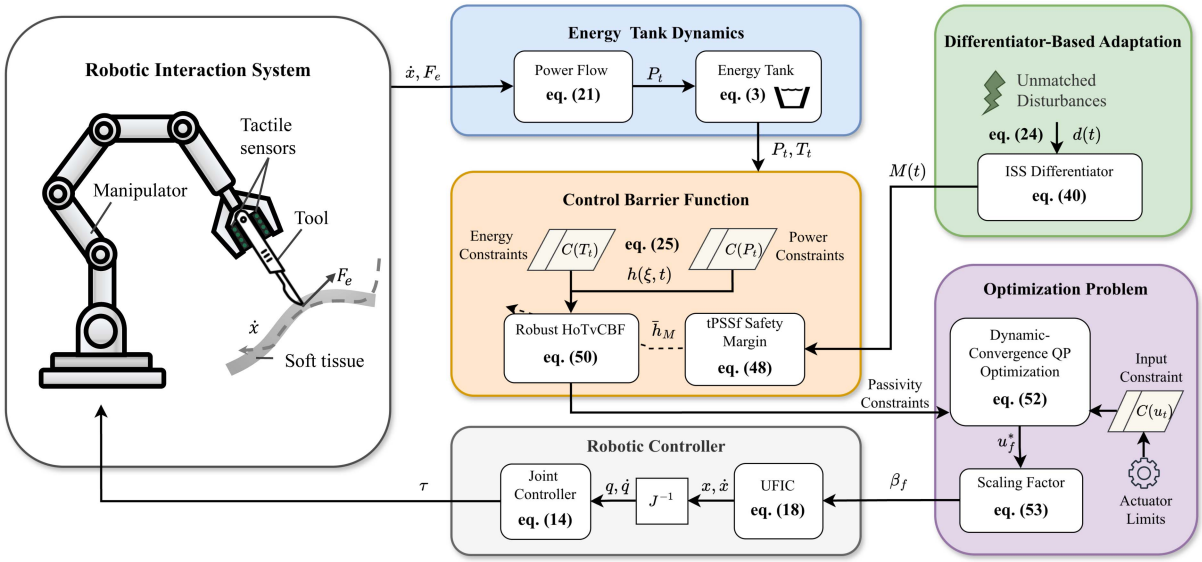


Fig. 2. Control scheme of the proposed approach for safe robotic cutting. Given the end-effector velocity $\dot{x}$ and contact force F_e from the robotic interaction system, the energy tank dynamics compute the power flow P_t and tank energy T_t to the CBF module. The ISS differentiator estimates unmatched disturbances $d(t)$ and provides the adaptive safety margin $\bar{e}(t)$ via tPSSf. The dynamic-convergence QP then generates optimal control input u_f^* under passivity and input constraints, and the robotic controller outputs the joint torque τ.

where the term $O(h(\xi,t))$ aggregates the remaining Lie derivatives along f and partial derivatives with respect to time t

$$O(h(\xi,t)) \triangleq \sum_{j=1}^{m-1} \left[L_f^j \left(\alpha_{m-j}(\psi_{m-j-1})\right)(\xi,t) + \frac{\partial^j \left(\alpha_{m-j}(\psi_{m-j-1})\right)(\xi,t)}{\partial t^j} \right]. \quad (31)$$

Any Lipschitz continuous controller satisfying (30) renders the time-varying safe set $\mathcal{C}(t) = \{\xi \in \mathbb{R}^n : h(\xi,t) \geq 0\}$ forward invariant.

Applying Definition 4 to constraints (25), the HoTvCBF conditions for power constraints are

$$\begin{aligned} k_f u - k_f \dot{e} - \dot{P}_{\min}(t) + \alpha_1(h_1(\xi,t)) &\geq 0 \\ -k_f u + k_f \dot{e} + \dot{P}_{\max}(t) + \alpha_2(h_2(\xi,t)) &\geq 0. \end{aligned} \quad (32)$$

For the energy constraints, define the sequences

$$\begin{aligned} \psi_{3,1}(\xi,t) &= \dot{e} - \dot{E}_{\min}(t) + \alpha_{3,1}(h_3(\xi,t)) \\ \psi_{4,1}(\xi,t) &= \dot{E}_{\max}(t) - \dot{e} + \alpha_{4,1}(h_4(\xi,t)) \end{aligned} \quad (33)$$

and the HoTvCBF conditions are as follows:

$$\begin{aligned} k_f u + (\alpha_{3,1} - k_f)\dot{e} - \ddot{E}_{\min}(t) - \alpha_{3,1}\dot{E}_{\min}(t) + \alpha_{3,2}(\psi_{3,1}) &\geq 0 \\ -k_f u + (k_f - \alpha_{4,1})\dot{e} + \ddot{E}_{\max}(t) + \alpha_{4,1}\dot{E}_{\max}(t) + \alpha_{4,2}(\psi_{4,1}) &\geq 0. \end{aligned} \quad (34)$$

Remark 2: The constraints in (32) and (34) guarantee safety only for the nominal system. They will fail in the presence of disturbances.

B. Time-Varying PSSf

For handling the uncertainties in relative-degree-one CBF systems, tPSSf has been proposed [27].

Definition 5 (Time-Varying PSSf [27]): Consider a disturbed system $\dot{\xi} = f + gu + d(t)$ and a safety function $h(\xi,t)$. Let the projected disturbance be $\delta(t,\xi) \triangleq \frac{\partial h}{\partial \xi} d(t)$, and assume a known, time-varying bound $\bar{\delta}(t)$ exists such that $|\delta(t,\xi)| \leq \bar{\delta}(t)$. A robust function $\bar{\psi}(\xi,t) \triangleq h(\xi,t) - \bar{\delta}(t)$ is defined. If an extended class-$\mathcal{K}$ function α exists satisfying the tPSSf dynamic condition

$$-\dot{\bar{\delta}}(t) + \bar{\delta}(t) \leq -\alpha(-\bar{\delta}(t)) \quad \forall t \geq 0 \quad (35)$$

then any controller u satisfying the constraint $L_f\bar{\psi} + L_g\bar{\psi}u + \frac{\partial \bar{\psi}}{\partial t} \geq -\alpha(\bar{\psi})$ renders the original set $\mathcal{C}(t) = \{\xi : h(\xi,t) \geq 0\}$ forward invariant.

The tPSSf cannot be directly applied to HoCBFs under unmatched disturbances. Consider the disturbed system from (24), which can be expressed in the general form

$$\dot{\xi} = f(\xi,t) + g(\xi,t)u + p(\xi,t)\delta_p(t) \quad (36)$$

where $p(\xi,t)$ represents the unmatched disturbance channel. Suppose this disturbance affects the energy constraint h_3 at the first relative degree. The HoCBF chain propagates this disturbance as follows.

1) First-order chain

$$\dot{\psi}_{3,1} = \dot{\psi}_{3,1}^{\text{nom}} + L_p\psi_{3,1}\delta_p(t) = \dot{\psi}_{3,1}^{\text{nom}} + \delta_1(\xi,t). \quad (37)$$

2) Second-order chain

$$\begin{aligned} \dot{\psi}_{3,2} &= \frac{d}{dt}(\psi_{3,1}^{\text{nom}} + \delta_1) \\ &= (\text{nominal dynamics}) + L_g\psi_{3,1}u + \dot{\delta}_1. \end{aligned} \quad (38)$$

The final mth order constraint depends not only on the disturbance δ_1 but also on its time derivative $\dot{\delta}_1$. Consequently, the final constraint takes the form

$$k_f u + \cdots + \alpha_{3,2}(\psi_{3,1}) + \underbrace{f_p(\delta_1, \dot{\delta}_1, \ldots)}_{\text{unknown terms}} \geq 0. \quad (39)$$

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Remark 3: The tPSSf requires knowledge of a disturbance bound $\bar{\delta}(t)$ and its derivative $\dot{\bar{\delta}}(t)$, which cannot accommodate these unknown disturbance derivatives that propagate through higher order barrier constraints.

C. Safety with Differentiator-Adaptive HoTvCBFs

The propagated disturbance vector, consisting of the unmatched disturbance and its derivatives up to the $(m-i)$th order, is treated as an unknown time-varying parameter vector denoted by $\theta(t) \in \mathbb{R}^q$. To estimate $\theta(t)$ in the presence of measurement noise, an ISS differentiator is employed.

Definition 6 (ISS differentiator [35]): Consider a noisy measurement $y(t) = \theta(t) + v(t)$, where $v(t)$ denotes bounded noise. An ISS differentiator is defined as a dynamical system $\dot{\hat{\theta}} = F(\hat{\theta}, y(t))$ that generates an estimate $\hat{\theta}(t) \in \mathbb{R}^q$ of the true parameter vector $\theta(t)$. The estimation error $e(t) \triangleq \theta(t) - \hat{\theta}(t)$ is uniformly bounded by a known, continuously differentiable time-varying function $\bar{e}(t) \geq 0$

$$\left\| \theta(t) - \hat{\theta}(t) \right\| \leq \bar{e}(t) \quad \forall t \geq 0. \tag{40}$$

According to the parameter estimate $\hat{\theta}$ and the error bound $\bar{e}(t)$, a robust high-order barrier function is constructed.

Theorem 1 (Differentiator-adaptive HoTvCBFs): Let $\tilde{\xi} \triangleq [\xi^T, \hat{\theta}^T]^T \in \mathcal{X} \times \mathbb{R}^q$ denote the state of the augmented system with dynamics $\dot{\tilde{\xi}} = \tilde{f}(\tilde{\xi}) + \tilde{g}(\tilde{\xi})u$. Let $\psi_{m-1}(\xi, \theta, t)$ be the final function in the HoCBF chain. It is assumed that ψ_{m-1} is Lipschitz continuous with respect to the parameter vector θ with a Lipschitz constant $L_h > 0$. A robust candidate function $\tilde{\psi} : \mathbb{R}_{\geq 0} \times \mathcal{X} \times \mathbb{R}^q \to \mathbb{R}$ is defined as

$$\tilde{\psi}(t, \tilde{\xi}) \triangleq \psi_{m-1}(\xi, \hat{\theta}, t) - L_h \bar{e}(t). \tag{41}$$

If a locally Lipschitz controller $u = k(\tilde{\xi}, t)$ exists satisfying

$$L_{\tilde{f}} \tilde{\psi}(\tilde{\xi}, t) + L_{\tilde{g}} \tilde{\psi}(\tilde{\xi}, t) u + \frac{\partial \tilde{\psi}}{\partial t} \geq -\alpha(\tilde{\psi}(\tilde{\xi}, t)) \tag{42}$$

where $\frac{\partial \tilde{\psi}}{\partial t}$ explicitly accounts for the term $-L_h \dot{\bar{e}}(t)$, then the original safety set $\mathcal{C}(t) = \{\xi \mid h(\xi, t) \geq 0\}$ is rendered forward invariant for the disturbed system.

Proof: First, the nonnegativity of the final HoCBF layer is established. Condition (42) renders the superlevel set of $\tilde{\psi}$ forward invariant. Assuming valid initial conditions $\tilde{\psi}(0, \tilde{\xi}(0)) \geq 0$, it follows that $\tilde{\psi}(t, \tilde{\xi}(t)) \geq 0$ for all $t \geq 0$. By substituting the definition from (41), the following is obtained:

$$\psi_{m-1}(\xi, \hat{\theta}, t) \geq L_h \bar{e}(t). \tag{43}$$

Leveraging the Lipschitz continuity of ψ_{m-1} with respect to θ and the ISS error bound, the inequality is derived as follows:

$$|\psi_{m-1}(\xi, \theta, t) - \psi_{m-1}(\xi, \hat{\theta}, t)| \leq L_h \left\| \theta - \hat{\theta} \right\| \leq L_h \bar{e}(t). \tag{44}$$

This inequality implies

$$\psi_{m-1}(\xi, \theta, t) \geq \psi_{m-1}(\xi, \hat{\theta}, t) - L_h \bar{e}(t). \tag{45}$$

Combining (43) and (45) yields $\psi_{m-1}(\xi, \theta, t) \geq 0$.

Next, the recursive property of HoCBFs is invoked. Since $\psi_{m-1}(\xi, \theta, t) \geq 0$, and by definition $\psi_{m-1} = \dot{\psi}_{m-2} + \alpha_{m-1}(\psi_{m-2})$, the inequality $\dot{\psi}_{m-2} \geq -\alpha_{m-1}(\psi_{m-2})$ holds. This condition ensures the forward invariance of the set $\{\xi \mid \psi_{m-2}(\xi, \theta, t) \geq 0\}$. By mathematical induction down to the zeroth layer, it is concluded that $\psi_0(\xi, t) = h(\xi, t) \geq 0$ for all $t \geq 0$. □

D. Time-Varying PSSf Differentiator-Adaptive HoTvCBFs

Although Theorem 1 compensates for unknown parameter derivatives via the error bound $\bar{e}(t)$, practical systems often encounter additional bounded disturbances $\tilde{d}(t)$, e.g., unmodeled dynamics. The dynamics of the fully disturbed augmented system are given by

$$\dot{\tilde{\xi}} = \tilde{f}(\tilde{\xi}, t) + \tilde{g}(\tilde{\xi}, t) u + \tilde{d}(t). \tag{46}$$

The safety verification problem is reformulated by utilizing $\tilde{\psi}(t, \tilde{\xi})$ as the candidate barrier function. Since $\tilde{\psi}$ exhibits relative degree one with respect to the control input u, this formulation constitutes a standard robust CBF problem amenable to the tPSSf approach. The projected disturbance acting on $\tilde{\psi}$ is defined as

$$\delta_{\text{proj}}(t, \tilde{\xi}) \triangleq \frac{\partial \tilde{\psi}}{\partial \tilde{\xi}} \tilde{d}(t). \tag{47}$$

It is assumed that a known time-varying bound $\bar{\delta}(t) \geq 0$ exists such that $|\delta_{\text{proj}}(t, \tilde{\xi})| \leq \bar{\delta}(t)$ for all $t \geq 0$. A unified robust safety function $\bar{\psi}$ is then constructed by incorporating both the parameter estimation error and the projected disturbance bound

$$\bar{\psi}(t, \tilde{\xi}) \triangleq \tilde{\psi}(t, \tilde{\xi}) - \bar{\delta}(t) = h_{\text{final}}(\tilde{\xi}, t) - L_h \bar{e}(t) - \bar{\delta}(t). \tag{48}$$

Theorem 2 (Time-varying PSSf differentiator-adaptive HoTvCBFs): Consider the DA-HoTvCBF function $\tilde{\psi}(t, \tilde{\xi})$ defined in Theorem 1 and the projected disturbance bound $\bar{\delta}(t)$. If an extended class-$\mathcal{K}$ function α exists such that the bound dynamics satisfy

$$-\dot{\bar{\delta}}(t) + \bar{\delta}(t) \leq -\alpha(-\bar{\delta}(t)) \quad \forall t \geq 0 \tag{49}$$

then any locally Lipschitz controller $u = k(\tilde{\xi}, t)$ satisfying

$$L_{\tilde{f}} \bar{\psi} + L_{\tilde{g}} \bar{\psi} u + \frac{\partial \bar{\psi}}{\partial t} \geq -\alpha(\bar{\psi}) \tag{50}$$

where $\frac{\partial \bar{\psi}}{\partial t}$ explicitly accounts for both $-L_h \dot{\bar{e}}(t)$ and $-\dot{\bar{\delta}}(t)$, renders the original safety set $\mathcal{C}(t) = \{\xi \mid h(\xi, t) \geq 0\}$ forward invariant for the fully disturbed augmented system.

Proof: The proof proceeds by establishing a chain of implications regarding the forward invariance of the associated superlevel sets. The robustness against the additive disturbance $\tilde{d}(t)$ is established initially. By the tPSSf theory [27], the satisfaction of condition (50) combined with the bound dynamics (49) guarantees that the set $\mathcal{C}_{\bar{h}} = \{\tilde{\xi} \mid \bar{\psi}(t, \tilde{\xi}) \geq 0\}$ is forward invariant. This implies that for any trajectory starting within $\mathcal{C}_{\bar{h}}$, the inequality $\bar{\psi}(t, \tilde{\xi}(t)) \geq 0$ holds for all $t \geq 0$.

Subsequently, substituting the definition of $\bar{\psi}$ from (48) yields the inequality $\tilde{\psi}(t, \tilde{\xi}) \geq \bar{\delta}(t) \geq 0$. Following the logic established in Theorem 1, the nonnegativity of ψ ensures that the

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true final-layer function satisfies $h_{\text{final}}(\xi, \theta, t) \geq 0$, despite the presence of parameter estimation errors.

The proof concludes by invoking the recursive structure of the HoCBF approach. The condition $h_{\text{final}} = \psi_{m-1} \geq 0$ implies the satisfaction of the lower layer barrier constraints, ultimately ensuring that $\psi_0 = h(\xi, t) \geq 0$. Thus, the original safety requirement is met. □

E. Optimization with Dynamic Convergence

Applying Theorem 2 to the passivity constraints in (25) yields the robust functions

$$\bar{\psi}_i(t, \tilde{\xi}) = \psi_{i,1}(\xi, \hat{\theta}, t) - L_h \bar{e}_i(t) - \bar{\delta}_i(t) \tag{51}$$

where indices $i \in \{1, 2\}$ correspond to power limits and $i \in \{3, 4\}$ to energy bounds.

To enforce these constraints while mitigating potential infeasibility arising from actuator limits or conflicting objectives, a dynamic-convergence QP is employed. This formulation relaxes the convergence rates ρ to maintain feasibility while minimizing deviations from the nominal control. The robust safety-critical controller is synthesized by solving

$$\begin{aligned} u^*, \rho^* = \underset{u \in \mathcal{U}, \rho \in \mathbb{R}^4}{\arg\min} \; & \frac{1}{2} \left\| u - u_{\text{nom}}(\tilde{\xi}, t) \right\|^2 + \sum_{i=1}^{4} c_{\rho_i} (\rho_i - \rho_0)^2 \\ \text{s.t.} \quad & L_{\tilde{f}} \bar{\psi}_i + L_{\tilde{g}} \bar{\psi}_i u + \frac{\partial \bar{\psi}_i}{\partial t} \geq -\rho_i \alpha_i(\bar{\psi}_i) \\ & (i = 1, \ldots, 4) \end{aligned} \tag{52}$$

where $\mathcal{U}$ denotes the actuator constraints and ρ_0 is the nominal decay rate.

Remark 4: Pointwise feasibility is guaranteed by formulation (52). The strict convexity of the objective function, combined with a nonempty feasible region (provided $\bar{\psi}_i > 0$), ensures the existence and uniqueness of the global optimal solution.

The optimal solution u_f^* yields the desired power flow satisfying all constraints, with scaling factor

$$\beta_f = \frac{u_f^*}{u_{\text{nom}}} = \frac{u_f^*}{P_t}. \tag{53}$$

V. Experiments

A. Experimental Setup

As shown in Fig. 3, the experimental platform integrates a ROKAE xMate3 Pro robot manipulator with a Robotiq 2F-85 gripper that holds a knife tool. To enable precise force sensing during tool-environment interaction, the gripper is equipped with two XELA 16-taxel uSkin tactile sensors. Each sensor contains a $m \times m$ $(m = 4)$ array of taxels mounted on the gripper fingertips, providing real-time external force feedback F_e at 100 Hz through a serial connection to a personal computer (PC). The system operates with the robot communicating state information, including joint positions, velocities, and torques $(q, \dot{q}, \tau)$ along with Cartesian position/velocity $(x, \dot{x})$, to the computing platform. The PC executes QP optimization

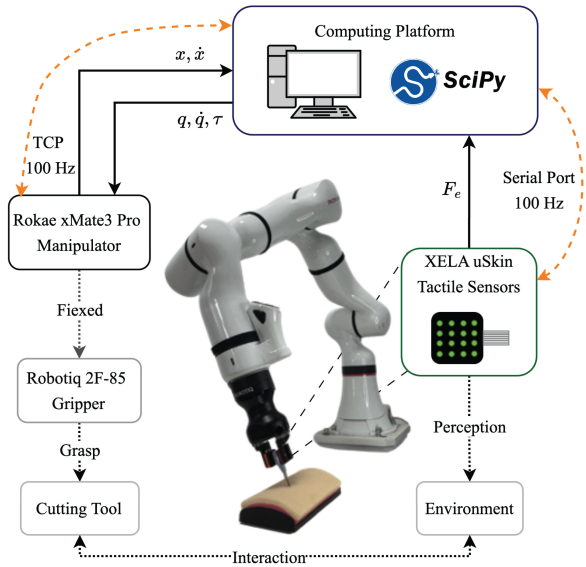


Fig. 3. Experimental platform for robotic cutting. The system consists of a ROKAE xMate3 Pro manipulator communicating with the computing platform via TCP/IP at 100 Hz, equipped with a Robotiq 2F-85 two-finger gripper and XELA uSkin tactile sensors providing force feedback at 100 Hz through a serial connection. The setup enables closed-loop control for tool-environment interaction during cutting tasks.

TABLE I
PARAMETERS IN THE EXPERIMENTS

Parameter Description	Symbol	Value
Impedance control gain	K_c	35
Force control gain	K_p	5
Force control gain	K_i	0.2
CBF control gain	k_f	50
CBF parameters for power	$p_{1,j}$	10
CBF parameters for energy	$p_{2,j}$	20

algorithms implemented in SciPy at 100 Hz and sends the control commands back to the robot.

To validate the proposed approach, four experimental scenarios are designed. First, the experiment implements through curved surface cutting with dynamic power constraint adjustment, which validates the ability of the proposed approach to adapt to time-varying safety requirements. Following this, the second experiment verifies the ability of the proposed approach to enforce both kinetic energy constraints and the resulting time-varying power bounds. The third scenario evaluates robustness under unmatched disturbances, specifically examining how the proposed approach maintains safety guarantees through adaptive safety margins. Lastly, a feasibility analysis experiment is conducted.

Table I presents the parameters used in the experimental setup. The impedance control gain K_c is tuned to provide appropriate compliance during tissue interaction while maintaining position accuracy. Force control gains (K_p, K_i) are selected to achieve stable force tracking with minimal steady-state error and overshoot during transitions between cutting phases. The CBF dynamics control gain k_f determines the convergence rate of the energy tank dynamics toward the desired power flow, with a theoretical lower bound $k_f > \frac{P_{\max}}{\lambda}$ ensuring passivity.

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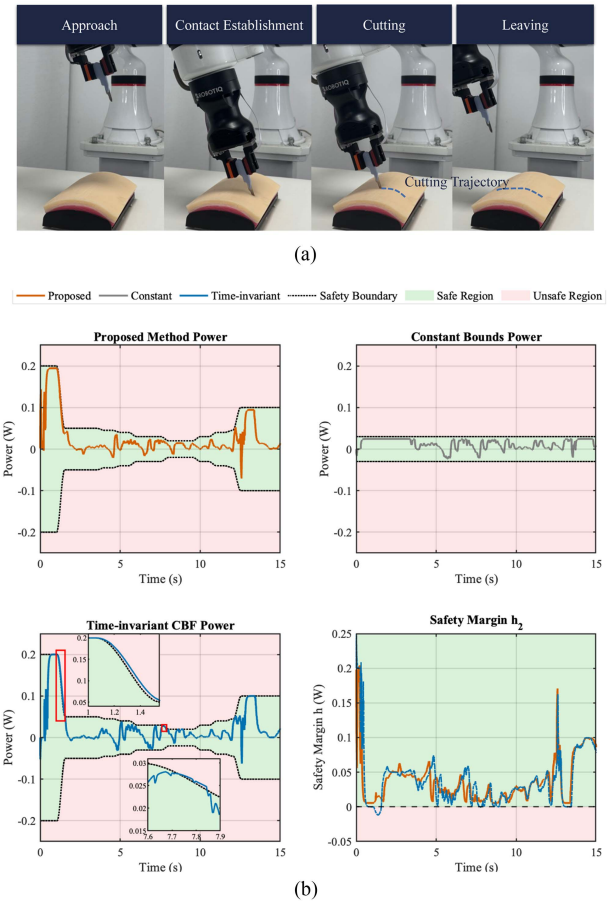


Fig. 4. Curved surface cutting experiment with time-varying power bounds. (a) Experimental snapshots showing four cutting phases: approach, contact establishment, cutting, and leaving. (b) Power response comparison under proposed approach, constant bounds approach, and time-invariant CBF approach, with safety margin $h_2(t)$ comparison between proposed approach and time-invariant CBF.

The HoTvCBF parameters $(p_{1,j}, p_{2,j})$ define the class-$\mathcal{K}$ functions that shape the convergence behavior toward the safe set boundaries for power and energy constraints, respectively.

B. Curved Surface Cutting with Time-Varying Power Bounds

The experiment investigates on a multistage cutting task with a curved surface geometry as illustrated in Fig. 4(a). The task consists of four distinct phases: approach, contact establishment, sustained cutting on a curved surface, and leaving. To capture local variations in tissue properties and geometry, each phase can be further subdivided into multiple segments.

The segmentation enables the implementation of segment-specific power constraints that adapt to the requirements of each portion of the cutting task. The power boundary learning mechanism updates these constraints through

$$ {}^{j+1}P_i^{\min/\max} = {}^{j}P_i^{\min/\max} - \kappa_f + {}^{j}\bar{\beta}_i(\kappa_f + \kappa_l) \tag{54}$$

with dynamic boundary evolution characterized by

$$ P(\phi) = \begin{cases} \Gamma(\phi), & \text{between phases} \\ P_i^{\min/\max}, & \phi \in \text{phase } i \end{cases} \tag{55}$$

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS

Metrics	Constant Bounds	Time-invariant CBF	Proposed
Force RMSE (N)	0.707 ± 0.053	0.749 ± 0.078	0.599 ± 0.041
Contact Reaching Time (s)	1.95 ± 0.21	1.05 ± 0.15	1.01 ± 0.12
Power Violations (%)	0.00 ± 0.00	4.11 ± 0.83	0.00 ± 0.00

where ${}^{j}P_i^{\min/\max}$ is the minimum/maximum power boundary for segment i in iteration j, κ_f, and κ_l are forgetting and learning coefficients, respectively, and ${}^{j}\bar{\beta}_i$ represents power utilization metrics from previous iterations. The phase indicator $\phi \in [0, 1]$ tracks progress through the cutting task, $P(\phi)$ is the power boundary function that adapts according to the current phase, and $\Gamma(\phi)$ is a smooth transition function, typically implemented as a sigmoid function, that ensures continuous boundary changes between phases.

The comparative results in Fig. 4(b) and Table II demonstrate performance differences between the three control strategies, namely, the CBF-Tank approach with constant bounds, time-invariant CBF-Tank approach, and the proposed approach. For the performance evaluation, three key metrics are utilized, including force RMSE quantifying force control precision, contact reaching time to measure efficiency during contact transition, and power violations, which evaluates safety constraint enforcement. Each control strategy has been tested in ten trials on the curved-surface cutting setup under identical conditions to ensure statistical validity of the results.

During variable force tracking scenarios, the proposed method achieves force RMSE of 0.599 N, reducing tracking error by 15.3% compared to the constant bounds approach at 0.707 N and a 20.0% improvement over the time-invariant CBF approach at 0.749 N. In terms of power boundaries, both the time-invariant CBF and the proposed approach use the time-varying segment-specific power constraints after learning process. The approaches with time-varying bounds complete the contact reaching phase in approximately 1.01–1.05 s, nearly 48% faster than the constant bounds method with 1.95 s. This efficiency improvement demonstrates the advantage of segment-specific boundaries over traditional fixed constraints, which is either overly conservative to ensure safety across all conditions or risk safety violations. Furthermore, the time-invariant CBF approach violates the power constraints during 4.11% of execution time, particularly during transient phases as highlighted in the red-boxed region in Fig. 4(b). These violations occur because the fixed barriers cannot adapt to rapidly changing task requirements. The proposed approach maintains complete safety with 0% power violations through incorporating boundary derivatives into constraint formulation.

C. Kinetic Energy Limitation

The second experiment evaluates the ability to maintain safety through dynamic kinetic energy and power constraints. During high-speed robotic operations, such as the approach phase in the cutting task, excessive kinetic energy risks safety to both the robotic system and nearby humans. To ensure safety during cutting operations, the kinetic energy of the manipulator is

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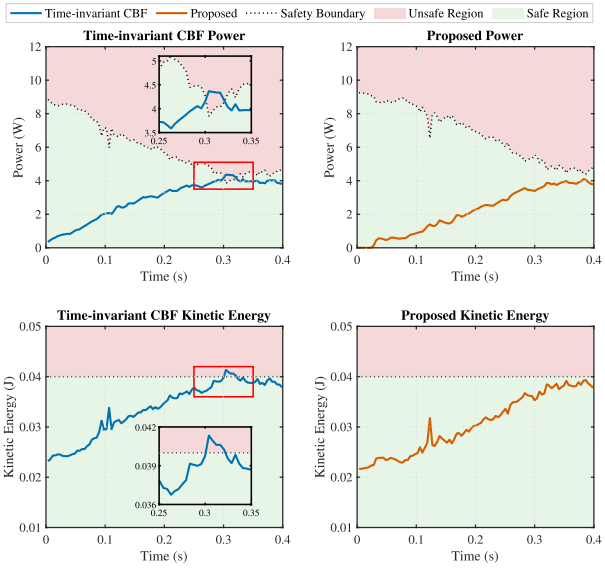


Fig. 5. Experimental results for the kinetic energy limitation experiment. Comparison of the time-invariant CBF and the proposed approach for maintaining time-varying power constraints and kinetic energy constraints during robotic motion.

governed by

$$E_{ke} = \frac{1}{2}\dot{x}^T M_c \dot{x} \tag{56}$$

which evolves according to

$$\dot{E}_{ke} = -P_t - \dot{\tilde{x}}^T (D_c \dot{\tilde{x}} + K_c \tilde{x} + F_e). \tag{57}$$

Using first-order approximation, the maximum kinetic energy constraint $E_{ke,k+1} \leq E_{ke,\max}$ introduces an additional time-varying power bound

$$P_{k\max}(t) = \frac{E_{ke,\max} - E_{ke,k}}{\Delta t} - \dot{\tilde{x}}^T (D_c \dot{\tilde{x}} + K_c \tilde{x} + F_e). \tag{58}$$

Based on this derivation, a time-varying power constraint is formulated that depends on the current kinetic energy state of the system, the maximum allowable kinetic energy, and other dynamic factors. If the kinetic energy of the manipulator is close to the safety limit, then the power budget reduces to prevent threshold violations.

The experiment evaluates the proposed approach and the time-invariant CBF approach using four metrics. Max power indicates the peak energy transfer rate during task execution, with lower values suggesting more conservative control. Power violations percentage indicates constraint enforcement reliability throughout the operation. Max kinetic energy quantifies the peak energy state of the system, directly correlating with potential collision severity. Kinetic energy violations percentage shows the effectiveness of preventing dangerous energy states during rapid movements.

As illustrated in Fig. 5 and Table III, the two approaches demonstrate different behaviors in managing power and kinetic energy constraints. The time-invariant CBF approach shows a steady increase in power consumption that eventually exceeds the boundary at approximately t=0.3 s, reaching a maximum of 4.366 W. This power boundary violation causes excessive energy accumulation, directly leading to kinetic energy exceeding the 0.04 J safety threshold. In contrast, the proposed method exhibits controlled power progression below the decreasing boundary limit, with 4.099 W maximum power. This results in kinetic energy approaching but never exceeding the safety threshold, maintaining 0.039 J maximum. Compared to the time-invariant CBF approach with a 4.13% violation rate of both power and kinetic energy constraints, the proposed approach maintains zero violations throughout the entire operation. Rather than reacting after the constraints are violated, the proposed approach adjusts before reaching safety constraints through, including boundary rate-of-change in the constraint calculations.

TABLE III
KINETIC ENERGY LIMITATION PERFORMANCE

Metrics	Time-invariant CBF	Proposed
Max Power (W)	4.366	4.099
Power Violations (%)	4.13	0
Max Kinetic Energy (J)	0.041	0.039
Kinetic Energy Violations (%)	4.13	0

D. Safety Performance Under Disturbances

To validate the robustness of the proposed approach, two representative disturbance scenarios are designed, including a contact-transition test and an external-disturbance test. The contact transition test examines the system response during the transition from noncontact to contact state, where sudden changes in system dynamics and unknown contact forces introduce significant disturbances. This scenario specifically validates the effectiveness of the ISS differentiator in tracking rapid dynamic variations, as the higher order derivatives required by HoTvCBF become particularly difficult to estimate during such transitions. The external disturbance test evaluates the system's ability to maintain safety under sudden perturbations during steady-state operation.

Two baseline methods are selected for comparison. The nominal controller, designed without explicit safety considerations, demonstrates the necessity of safety-critical control by showing the consequences of ignoring disturbance effects. The PSSf with fixed bound method employs a constant worst-case disturbance bound $\bar{\delta} \geq \sup_{t\geq 0} |\delta(t)|$ to guarantee safety, representing the conventional robust CBF approach. The comparison between this fixed-bound method and the proposed approach highlights the advantage of adaptive disturbance bound estimation over conservative static bounds.

Fig. 6 illustrates the dynamic response of the safety function $h_2(t)$ under both scenarios. In the contact transition test, when contact occurs at approximately $t = 1$ s, the nominal controller responds slowly and allows $h_2(t)$ to drop deeply into the unsafe region before eventually recovering. The PSSf with fixed bound maintains a consistently large safety margin throughout the process, reflecting its conservative nature. The proposed method demonstrates adaptive behavior. During the disturbance, $h_2(t)$ decreases toward the boundary but remains positive, then recovers as the ISS differentiator tracks the changing dynamics. In the external disturbance test, a similar pattern emerges at $t = 4.8$ s,

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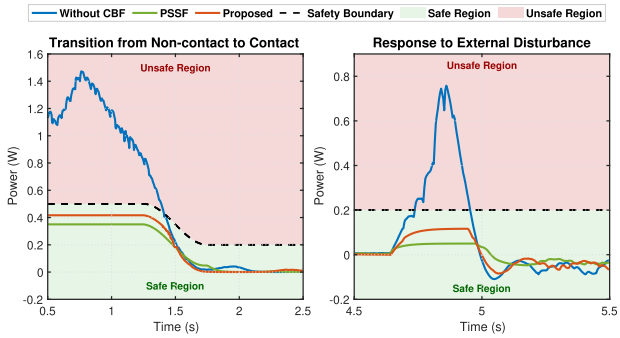


Fig. 6. Experimental validation of the robustness of the proposed tPSSf-DA-HoTvCBF method. (Left) Contact transition test: system response during transition from noncontact to contact state. (Right) External disturbance test: safety performance under sudden external perturbation. The proposed method maintains safety constraints but the nominal controller violates the safety boundary. The fixed-bound PSSf remains safe but exhibits conservative behavior.

TABLE IV
QUANTITATIVE SAFETY PERFORMANCE COMPARISON UNDER DIFFERENT DISTURBANCE SCENARIOS (MEAN OVER 10 TRIALS)

Method	Contact transition		External Disturbance	
	Violation	Min $h_2(t)$	Violation	Min $h_2(t)$
Nominal Controller	YES	-0.9732	YES	-0.5578
PSSf (Fixed Bound)	No	0.1500	No	0.1500
Proposed Method	**No**	**0.0833**	**No**	**0.0836**

where the proposed method shows rapid response and recovery while the fixed-bound approach maintains an unnecessarily large margin even when disturbances are minimal.

The quantitative results in Table IV summarize the safety performance over ten repeated trials. The nominal controller violates safety in both scenarios, with a more severe violation during contact transition at $\min h_2(t) = -0.9732$ compared to -0.5578 under external disturbance, indicating that transient dynamic changes pose greater challenges. Both safety-critical methods guarantee $h_2(t) \geq 0$, but exhibit different levels of conservatism. The fixed-bound PSSf maintains $\min h_2(t) = 0.1500$ in both scenarios due to its constant worst-case bound assumption, while the proposed approach achieves $\min h_2(t) = 0.0833$, representing approximately 44% reduction in safety margin. This reduction translates to improved task performance, as the system operates closer to its physical limits without compromising safety guarantees.

E. Feasibility Analysis

This experiment is designed to analyze QP solving feasibility, particularly in cases where the phase-specific time-varying power boundaries might conflict with the input constraints. The experimental setup focuses on the power boundary contraction during the transition from approach to contact establishment. The study compares the QP solver failure rate and the satisfaction rate of each constraint across the total solution time under two distinct methods: with dynamic convergence enabled versus disabled.

As shown in Table V, without dynamic convergence, the QP solver exhibits a 2.72% failure rate, with constraint violations primarily occurring in the power upper bound with 97.28% satisfaction and β upper bound with 97.96% satisfaction. In contrast, the dynamic-convergence on implementation achieves 100% satisfaction across all constraints. Under identical initial conditions, the proposed approach implements constraint relaxation, allowing the controller to maintain feasibility throughout the experiment.

TABLE V
COMPARISON OF CONTROL PERFORMANCE WITH AND WITHOUT DYNAMIC CONVERGENCE

Metric	Dynamic-convergence Off (%)	Dynamic-convergence On (%)
QP Failure Rate	2.72	0.00
Power $h_1 \geq 0$	100.00	100.00
Power $h_2 \geq 0$	97.28	100.00
Energy $h_3 \geq 0$	100.00	100.00
Energy $h_4 \geq 0$	100.00	100.00
Input $u \geq u_{min}$	99.90	100.00
Input $u \leq u_{max}$	97.96	100.00

VI. CONCLUSION

This article presents an energy tank management approach with tPSSf-DA–HoTvCBF for robust safe robotic cutting. For cutting tasks, the proposed approach conducts smooth regulation of phase-specific time-varying energy and power constraints throughout different cutting phases, preventing rapid energy release during unexpected motion. To address unknown higher order derivatives in HoTvCBF, an ISS differentiator is employed to estimate time-varying parameters and tPSSf is utilized to adaptively bound estimation errors. Compared to fixed-bound robust CBF methods, this approach achieves safety guarantees with reduced conservatism, which is validated through contact transition and external disturbance experiments. In addition, the dynamic-convergence HoTvCBF-QP solver resolves conflicts between passivity requirements and input constraints, maintaining feasibility of the optimization problem. In future work, the proposed approach can be extended to human–robot interaction and multirobot collaborative systems.

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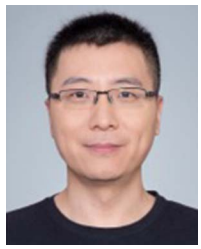
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